Worksheet

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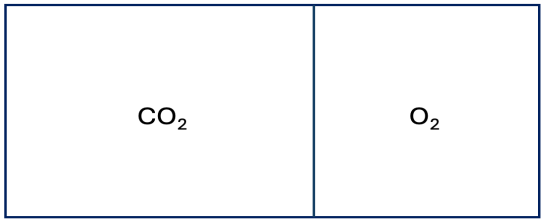
1 Ideal Gas Law - Practice Problem

(a) If 3 moles of CO_2 is kept in 100° C in a chamber of 6 L, then what is the pressure of the gas?

Optional: Using any programming language/ graphing tool, plot how the pressure (in SI units) of this gas varies if the temperature is varied from $-25°$ C to 225° C.

(b) If 2 moles of O_2 is kept in the same pressure in 3 L chamber, then what would be the temperature of the gas?

(c)



Now imagine a set-up like this. We have our 3 moles of CO_2 with 6 L volume in the left compartment, and 2 moles of O_2 with 3 L volume in the right compartment. The whole system is maintained at 25°C and the partition between the two compartments is sliding.

i) Calculate the pressures for both the gases.
ii) What is the pressure difference?

iii) This pressure difference would make the partition move. If the area of the partition is 750 cm^2, then what is the force on the partition?
iv) If the mass of the sliding partition is 25 g, then what is the acceleration of the partition?

d) For next week, we are going to learn about radiation pressure. Check out the resources in this link - https://byjus.com/physics/radiation-pressure/ , and read the material and contact me if you have any questions.